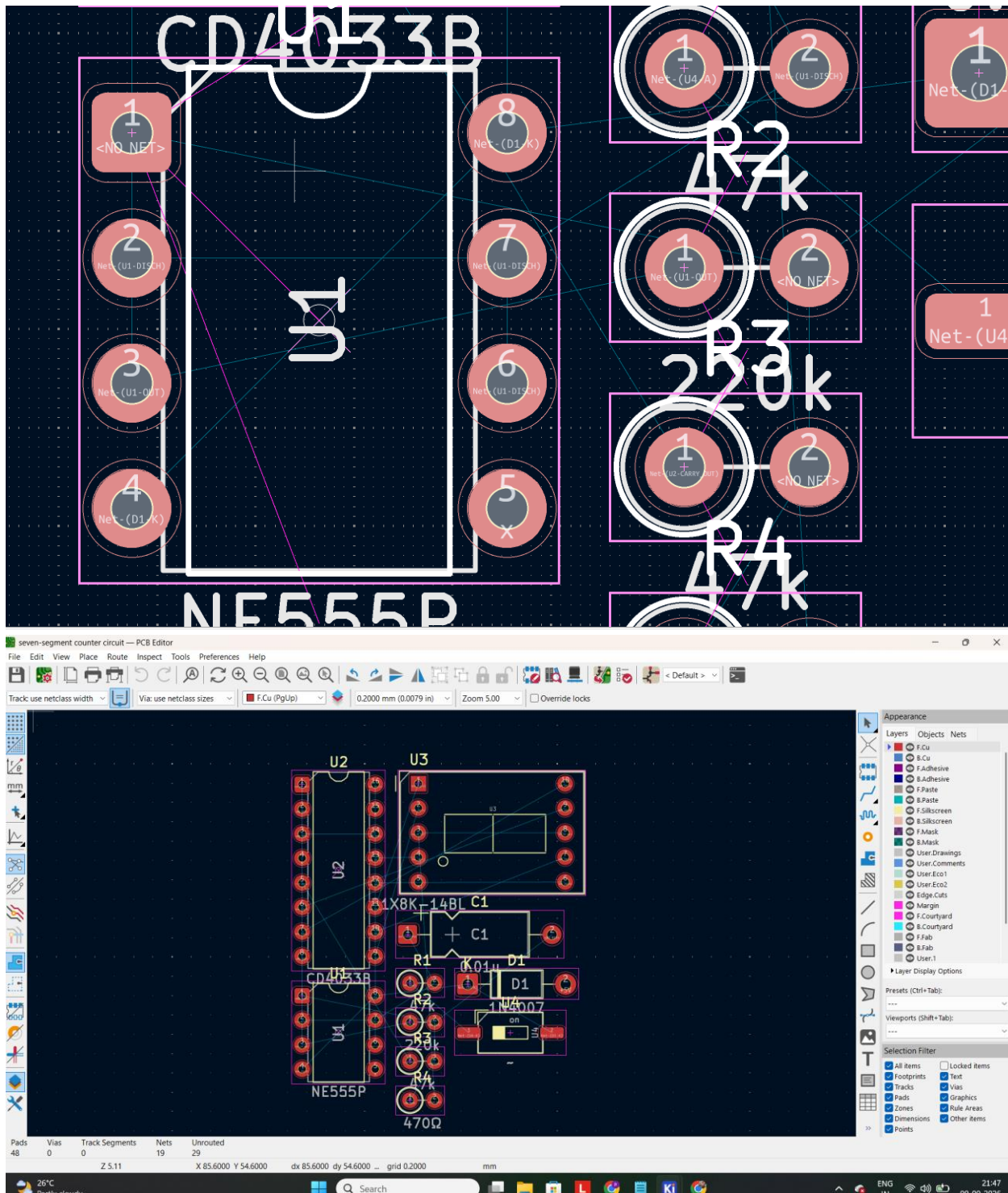






REFLECTION:

Correct footprint mapping and proper PCB rule setup are critical for ensuring that a PCB

can be manufactured and assembled successfully. A footprint defines the physical size, pad arrangement, hole dimensions, and placement of a component on the PCB. If the wrong footprint is assigned, the component may not fit on the board, its pins may not align with the pads, or it may be impossible to solder correctly. This can lead to assembly failures, rework, and additional manufacturing costs.

PCB rule setup is equally important because it defines requirements such as track width, clearance, via sizes, and design constraints. These rules help prevent electrical and manufacturing problems during PCB fabrication. If the rules are not configured correctly, tracks may be too close together, causing shorts, or too narrow to safely carry the required current.

Overlooking footprint mapping and PCB rules can result in design errors that may only be discovered after manufacturing. This can cause delays, increased costs, board failures, and even safety issues. Therefore, carefully verifying footprints and running ERC/DRC checks before manufacturing is essential for producing a reliable and manufacturable PCB.